



"Digital Dreams Unlocked"

Audit and Compliance Management

Pharmix AI

Empowering Pharma through Digital Insights

Wednesday, December 17,
2025

Pharmix AI : All Time Readiness

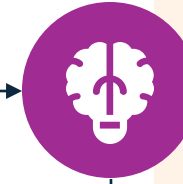
01 Ensuring Regulatory Compliance

Identification of the GAPS wrt to regulatory guidelines to ensure compliance.

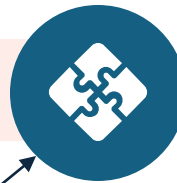


02 Identifying & Mitigating Risks

Assessing & providing recommendations to address the GAPS to maintain the state of Compliance.



Continued State of Compliance



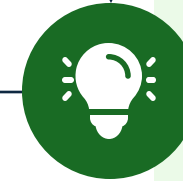
04 Continuous Improvement

Facilitating ongoing monitoring to drive continuous improvements.



03 Improving Quality Management

Evaluating and enhancing the effectiveness of quality management systems.



Pharmix AI – Smart Audit Management Solution

Challenge Statement: Traditional Audit Process (Manual or Electronic), are Error Prone, biased and Time consuming hurting compliance and Efficiency. This leads to regulatory risks and higher Operational Cost.



Focuses Audit efforts on high risk areas using AI driven Intelligence, optimizing resource allocation.

Risk-Based Auditing



Maintain compliance status, ensuring readiness for any audit and any time.

Audit-Ready Workflows



Automatically aligns & updates regulatory changes for ensuring seamless compliance.

Pharma Hook

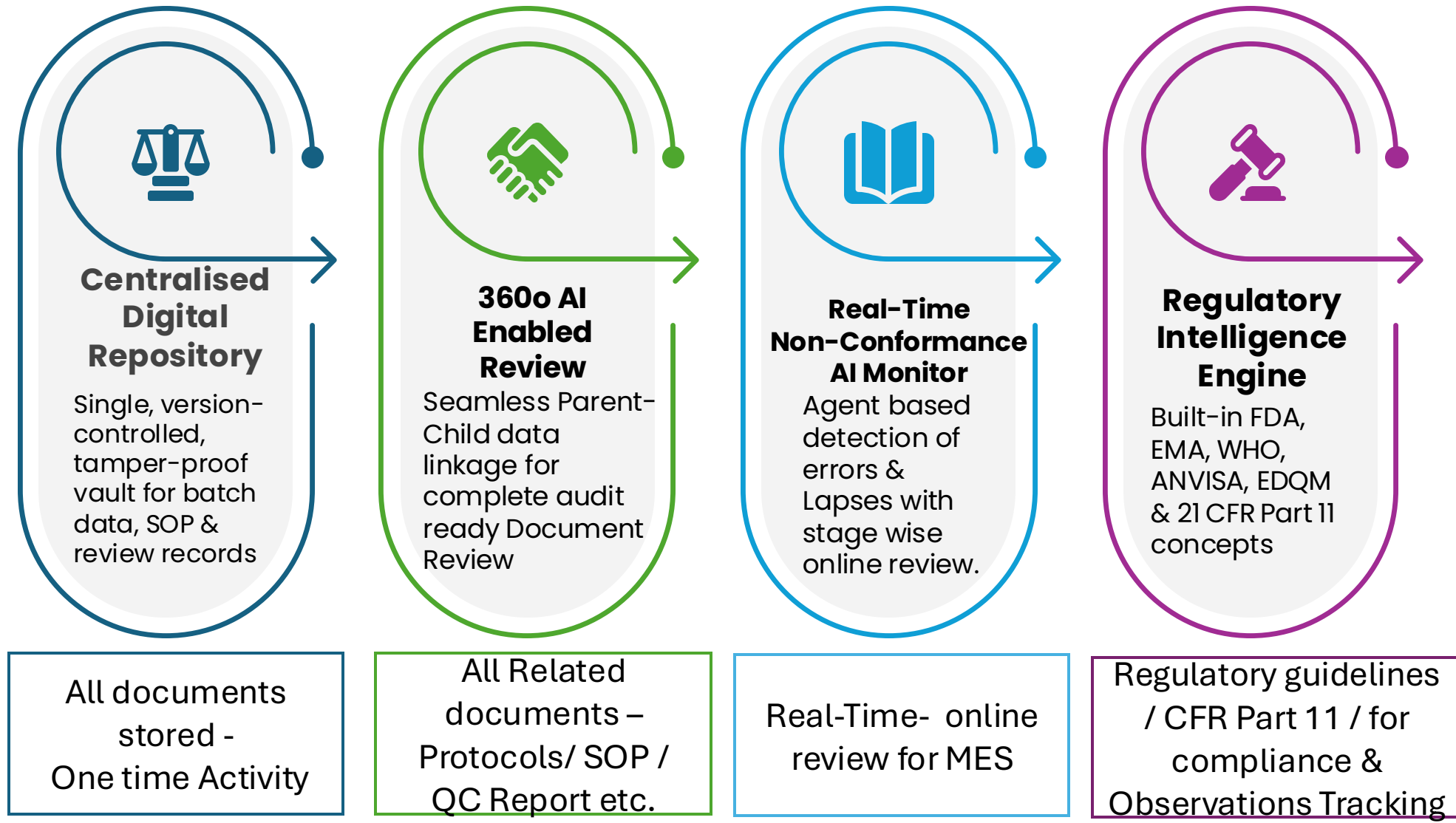


Provides an instant Compliance score card for Realtime performance monitoring

Compliance Scoring

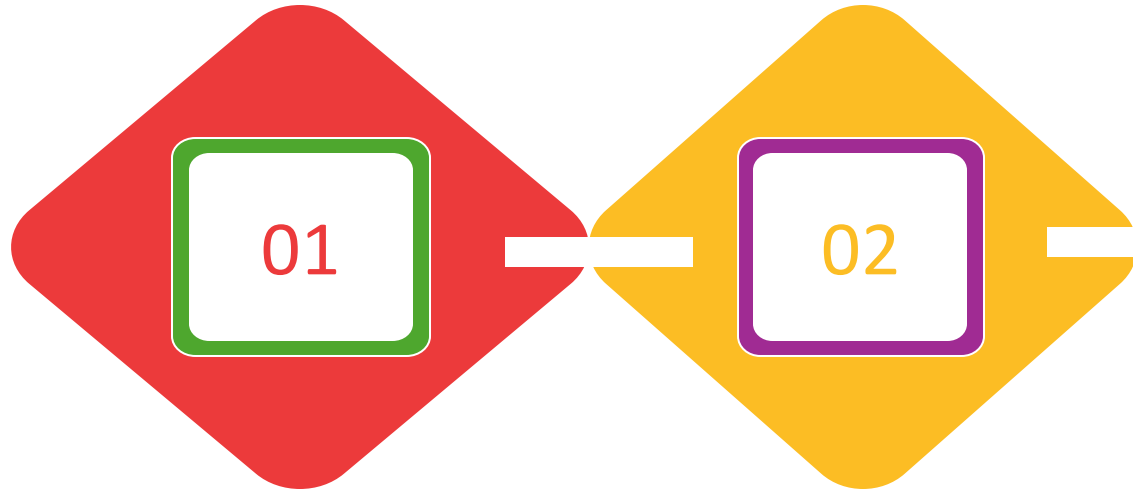
Compliance Management & Review – Digital Transformation Initiative

Problem Statement: Ensuring reliable, audit ready data lineage across entire product lifecycle.



Compliance Task Management

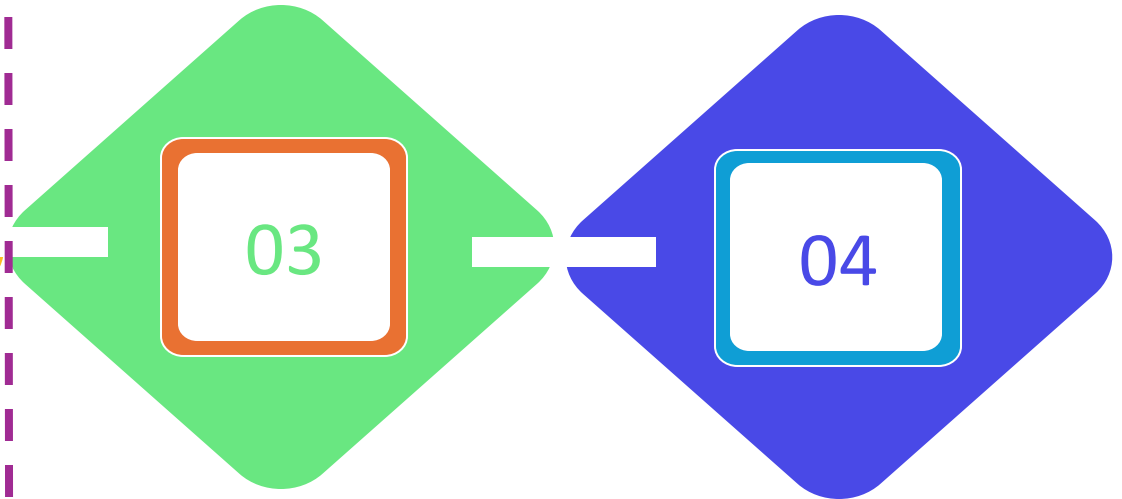
One Time Activity- 01 & 02
Master Configuration: ~3 min



**Inhouse Checklist
Configuration**

**Inhouse SOP
Configuration**

Checklist, ALCOA+, GMP & SOP- 03 & 04
Review Reports: ~ 5 Min



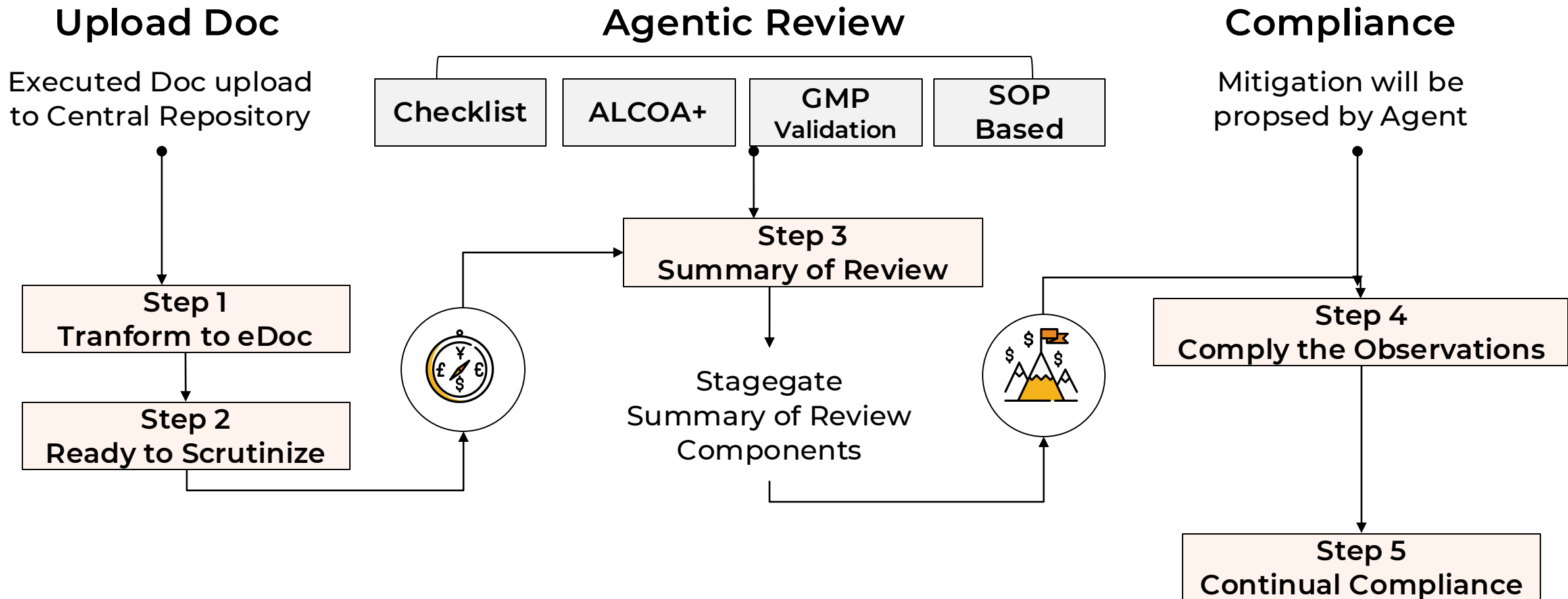
**Upload BMR /
Document**

**Review Report ready in
~10 Mins**

Where to Use:

1. For routine batches
2. Scale Up, Exhibit, regulatory or customer audits where **holistic review** is required – **PARS**

Agentic Review : Architecture



Application Review Capabilities

Compliance Verification Capabilities

Custom Checklist Review-
Configurable Checklist per need.



ALCOA / GDP Verification
Default checks against principles



SOP - based GAP Assessment
SOP Vs BMR Clause comparison.



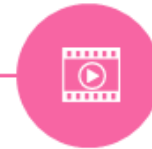
GMP Rule Verification-
System generated Default rules



Intelligent Review Capabilities



Logical Review -
Compare with reference documents



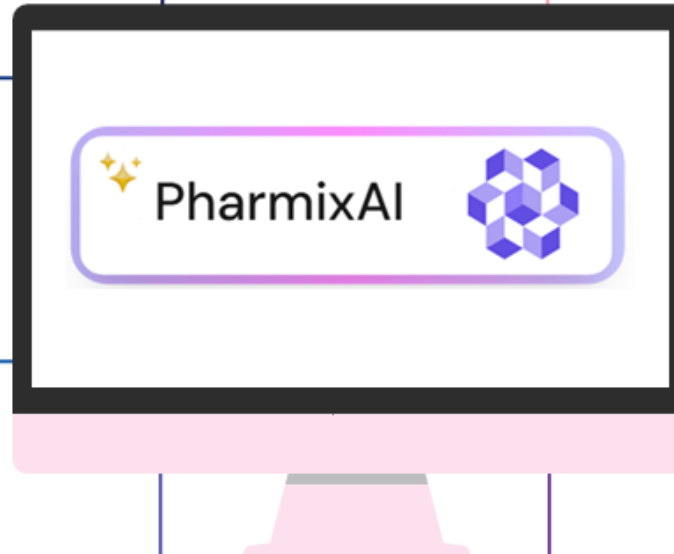
Interpretation Ability-
Dataset Interpretation.



Parent to Child Comparison-
Verify document lineage



Risk Flaging with mitigation-
Action and Attention flags.



Quantitative Insights

DYNAMIC DASHBOARDS

SEE BEYOND THE NUMBERS

- Realtime Document Compliance Insight.
- Product Wise Risk level Plot help to improve product Efficiency.
- Helps to Improve Overall Operational Efficiency

Overview

My Tasks ↗

This week ▾

32

Over Due for Approval

This week ▾

0

At Risk Documents

This week ▾

45

Total Documents

This week ▾

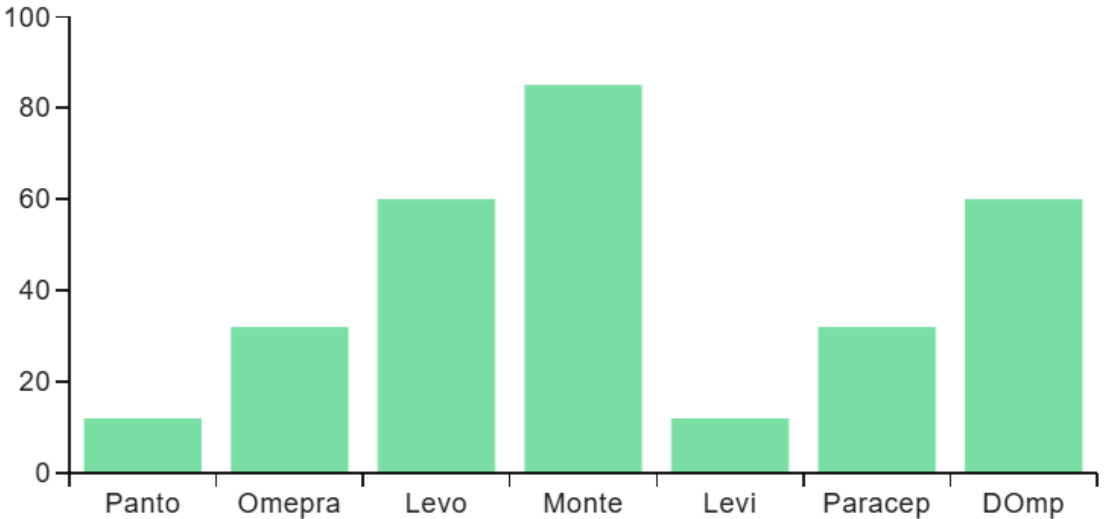
102



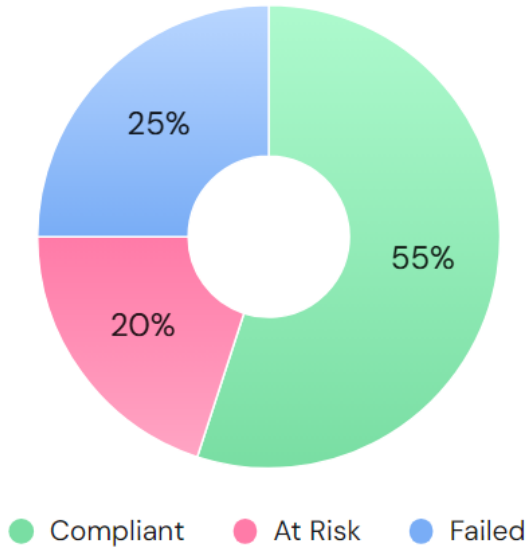
Risk Level Plot



Search Product



Overall Compliance Status Break down



Quantitative Insights

DYNAMIC DASHBOARDS

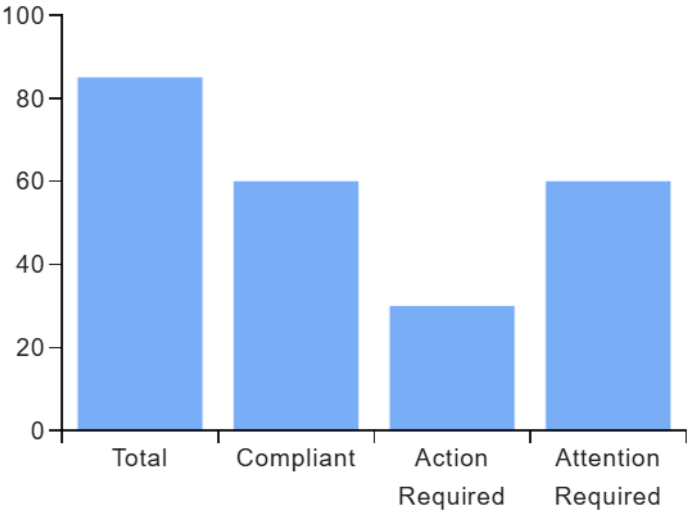
SEE BEYOND THE NUMBERS



Product Wise Compliance Status

Q Search Product

Levocet 5 mg, B.No.: AB2104



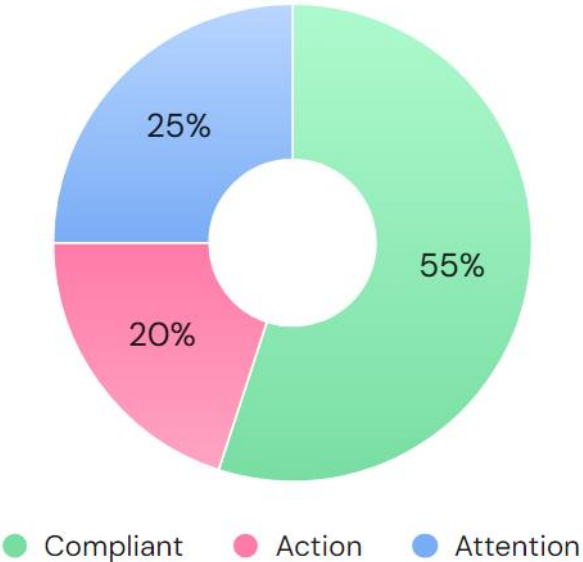
Section wise Compliance Breakdown

Q Search Section

Q Search Product

Checklist Review

Levocet 5 mg, B.No.: AB2104



Quantitative Insights

BATCH COMPARISON SPOTTING THE VARIATION

- **AI INSIGHT** : AI findings and Recommendations help to maintain the product compliance score.
- **Real-Time Visualization**: Move beyond static reports with a dynamic dashboard that visualizes current and historical batch data instantly
- **Quantitative Performance**: Clearly display Batch-to-Batch Variation for all Critical Quality Attributes and Critical Process Parameters
- **Trend Identification**: Plot Control Charts (e.g., Shewhart Charts) to detect non-random trends, shifts, or drifts

BATCH COMPARISON SPOTTING THE VARIATION

Batch Comparison Dashboard

Back to Form

Product: Product A Batches: Batch 001, Batch 002, Batch 003 Date Range: 1-11-2025 – 16-12-2025

Batch Comparison Report

Batch A1

98.2%

Status: 0

Compliance-Score: 94

Batch A2

95.5%

Deviations: 2

Compliance-Score: 88

Batch A3

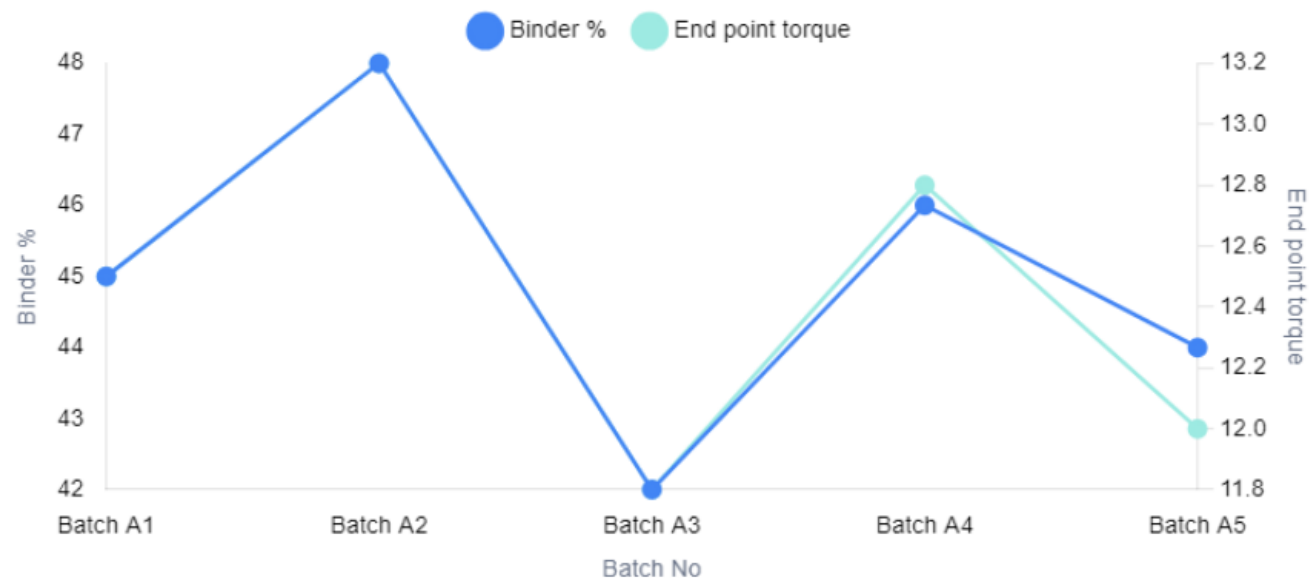
92.1%

Fatal: 5

Compliance-Score: 71

Quantitative Insights

Granulation Comparison (AI Insights)



Batch A3 shows abnormal granulation torque = 19.6 better than batch par correlating with friability failure.



AI insight: A2 under-blended by 3 minutes – Higher CU variation.

AI Findings & Recommendations

AI Summary

- Parameter deviation in 3 stages
- 2 parameters outside allowed ranges
- 1 recurring deviation pattern detected

AI Recommended Actions

- Optimize granulation torque target + 0.822 Nm
- Reduce turret speed by 3 min for higher CU variation
- Add IPC sampling at 20 min interval

Process Findings

- ⚠ Missing signatures! Batch A3
- ⚠ Cleaning log mismatch – Batch A2
- ⚠ Manual entry one write flagged

Generate CAPA Report →

Regulatory Intelligence

Regulatory Intelligence Report

1. Report Header

Report Title: ALCOA+ Validation Report:
Product: 1032500714
Batch No: 1032500714
Document: PROD/BMS/2294R2
Validation Section: Complete BMR Audit

This section provides a thematic cross-reference between identified non-compliances in this report and publicly reported FDA 483 / Warning Letter observations for other companies. It does not imply that such letters were issued for this specific batch or facility.

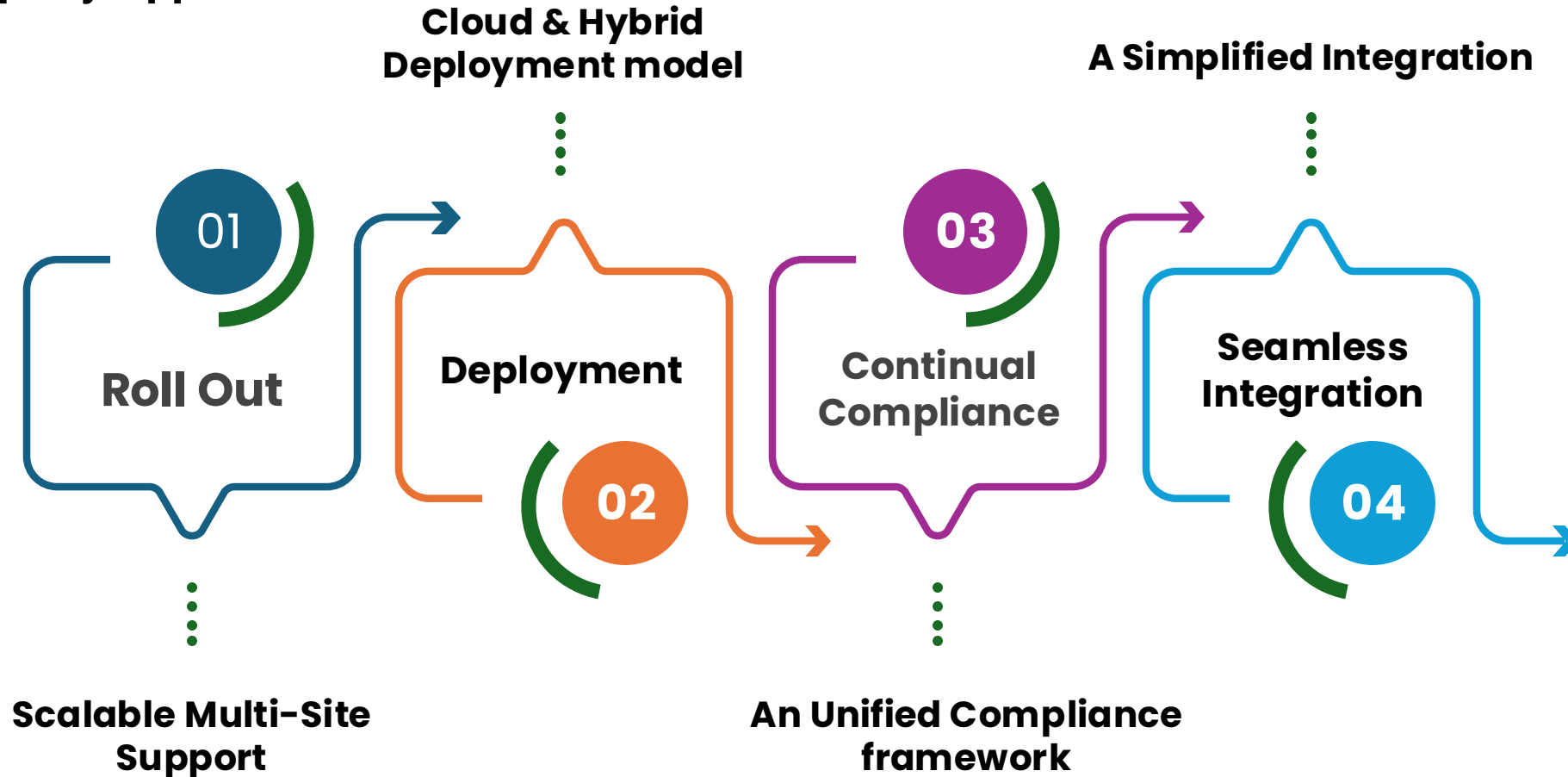
Internal Non-Compliance / Gap	Related Section in This Report	Relevant FDA CGMP Citation (21 CFR)	Example FDA 483 / Warning-Letter Observation (Company / Theme)	Year	Date
Missing QA signature for final yield verification (Page 4) – critical yield calculation recorded but QA verification field blank.	Executive Summary – Attributable Validation Summary Note 1.	21 CFR 211.188 – Batch production and control records must include documentation that each significant step (including yield) is performed and checked. 21 CFR 211.192 – Production records must be reviewed and approved by the quality unit before batch release; discrepancies must be investigated.	Example (FDA Form 483 – Right Value Drug Stores, LLC) – FDA cited the firm for failure to thoroughly review discrepancies in a sterile injectable batch, under 21 CFR 211.192, during an inspection leading to a 483 issued to Right Value Drug Stores, LLC in Irving, TX.	2025	2025-07-18 (Form 483 issue date)

Pharmix AI: Time, Quality, and ROI Comparison

Focus Area	Metrics	Manual Review	Pharmix AI Review
Time	Review Time per BMR	24 - 48 Hours (One Reviewer from PRD & QA)	10 Minutes (~99 % Time Reduction)
Quality Shield	De-Risking Batch Release	~20 (Limited Checkpoints, Latency)	~150 for said BMRs (Exhaustive, End-to-End)
Efficiency	Review Scope	Chances of Human errors (Fatigue/Bias)	Holistic & Concurrent (Check of every data point)
Quality Risk	21 CFR § 211.192	Review Oversight (OOS in pH))	Detects (Flags the OOS result)
	21 CFR Part 11 11.50, 11.200 (a) ER and ES	Review Oversight (Meta Data Verification viz. Balance Prints, Machine Printouts etc.)	Flags (Identified the discrepancy)

Integration & Scalability

Problem Statement: Need of a system that can easily scale from 10 users to 1,000, and integrate with existing 3rd party Application





Thank you!

Do you have any questions?